Skills Summary

Interior and Exterior Polygon Angle Sums

Use this reference while completing your worksheet or when studying.

Skill 1 — Interior angle sum of a convex polygon

Theorem: For a convex polygon with n sides, sum of interior angles $= (n-2)\,180°$.

- *Why:* diagonals from one vertex cut the polygon into exactly $n-2$ triangles, and each triangle holds $180°$.
- Running it backwards finds n: set $(n-2)180$ equal to the given sum.
- If one angle is missing, add the known angles and subtract from the sum.

Example: Find the interior angle sum of a convex heptagon.
Step 1. Only the number of sides is known and a total is wanted, so this is the interior angle sum theorem, not the exterior one.
Step 2. $n = 7$, so the sum is $(7-2)180° = 5 \cdot 180°$. $\Rightarrow$ **900°**

Try it: Three angles of a convex quadrilateral are $88°$, $154°$, and $97°$. Find the fourth, x.

check: $\boldsymbol{x = 21°}$

Skill 2 — Exterior angles always total 360 degrees

Theorem: Taking one exterior angle at each vertex, sum of exterior angles $= 360°$ — for *every* convex polygon, whatever n is.

- At each vertex the interior and exterior angles form a linear pair: interior + exterior = $180°$.
- Regular polygon: one exterior angle $= \dfrac{360°}{n}$, so $n = \dfrac{360°}{\text{one exterior angle}}$.

Example: Find one exterior angle of a regular nonagon.
Step 1. The polygon is regular and an exterior angle is wanted, so share the fixed $360°$ total equally among the vertices.
Step 2. $n = 9$, so one exterior angle is $360° \div 9$. $\Rightarrow$ **40°**

Try it: Each exterior angle of a regular polygon is $45°$. How many sides has it?

check: $\boldsymbol{n = 8}$

Skill 3 — Angles of a regular polygon

Formulas: In a *regular* n-gon all angles are congruent, so

$$\text{one interior angle} = \frac{(n-2)\,180^\circ}{n}, \qquad \text{one exterior angle} = \frac{360^\circ}{n}.$$

The two are supplementary, so either one gives the other instantly.

Example: Find one interior angle of a regular pentagon.

Step 1. Regular means the interior total is shared equally, so divide the interior angle sum by the number of angles.

Step 2. $\dfrac{(5-2)180^\circ}{5} = \dfrac{540^\circ}{5}$. $\Rightarrow$ $\mathbf{108^\circ}$

Try it: Each interior angle of a regular polygon is 140°. How many sides has it?

check: $\boldsymbol{n = 9}$

(!) Watch out: The 360° belongs to the *exterior* angles only. Dividing 360° by n gives an exterior angle, never an interior one — and $(n-2)180^\circ$ is a total for the whole polygon, not a single angle.